TECHNICAL REPORT

Multi-Object Detection and Persistent ID Tracking in Public Sports Footage

1. Objective

The objective of this assignment is to design and implement a practical computer vision system capable of detecting multiple moving players in a publicly accessible sports video and assigning persistent identities to those players across consecutive video frames. The final output should visually demonstrate stable player tracking using annotated bounding boxes and unique ID labels.

To accomplish this, an end-to-end multi-object tracking pipeline was developed using a pretrained object detector and an appearance-assisted tracking algorithm.

2. Public Video Source and Segment Selection

For this assignment, a publicly accessible football tactical camera broadcast video available on YouTube was selected as the source input.

Source Video Link:

https://youtu.be/M9mKnmt0YaM?si=MmPbVWQH2FzXQRME

Processed Video Segment:

00:00:45 to 00:00:58

A 13-second representative segment was intentionally extracted from the original public source because it contains multiple simultaneously visible players, continuous player motion, and relatively stable camera panning. This makes the selected clip suitable for evaluating practical persistent multi-object tracking behavior.

3. System Pipeline Design

The implemented tracking system follows a sequential frame-processing pipeline:

Public Video Input → Frame Extraction → Human Detection → Multi-Object Tracking → Persistent ID Mapping → Frame Annotation → Final MP4 Output

First, the selected football clip is read frame-by-frame using OpenCV. For each frame, all visible human subjects are detected using a pretrained object detection model. These detections are then passed to a tracking algorithm that attempts to preserve the same identity labels over time. Finally, annotated frames are written back into a new processed output video.

4. Object Detection Model Used

YOLOv8s from the Ultralytics framework was selected as the human object detector.

This model was chosen because:

• it provides strong person detection accuracy,

• it performs inference efficiently on a local machine,

• it is lightweight but more reliable than smaller nano variants,

• it integrates smoothly with tracking frameworks.

For this assignment, only the “person” class was retained from the detector output so that all visible football players and referees become valid tracking subjects while irrelevant field objects are ignored.

5. Persistent Tracking Algorithm Used

To maintain identity continuity across frames, the BoT-SORT tracking algorithm was used.

BoT-SORT is a practical multi-object tracking algorithm that combines:

• motion prediction,

• Kalman-based trajectory estimation,

• detection-to-track association,

• appearance-assisted identity matching.

Compared to basic motion-only trackers, BoT-SORT provides improved identity preservation when players move quickly, overlap, or temporarily leave ideal detection positions.

The tracker internally generates raw tracking IDs. For better readability in the final annotated video, these raw IDs were remapped into simplified human-readable labels such as Player 1, Player 2, Player 3, and so on.

6. Visualization and Output Video Generation

For every detected and tracked player, the following visual annotations were added on each frame:

• bounding rectangle around the player,

• unique player identity label,

• center movement point indicating current position.

Additionally, the total number of tracked players visible in the current frame is displayed at the top-left corner.

After annotation, all processed frames are sequentially reassembled and exported as a final MP4 output video, providing a clear visual demonstration of persistent player tracking.

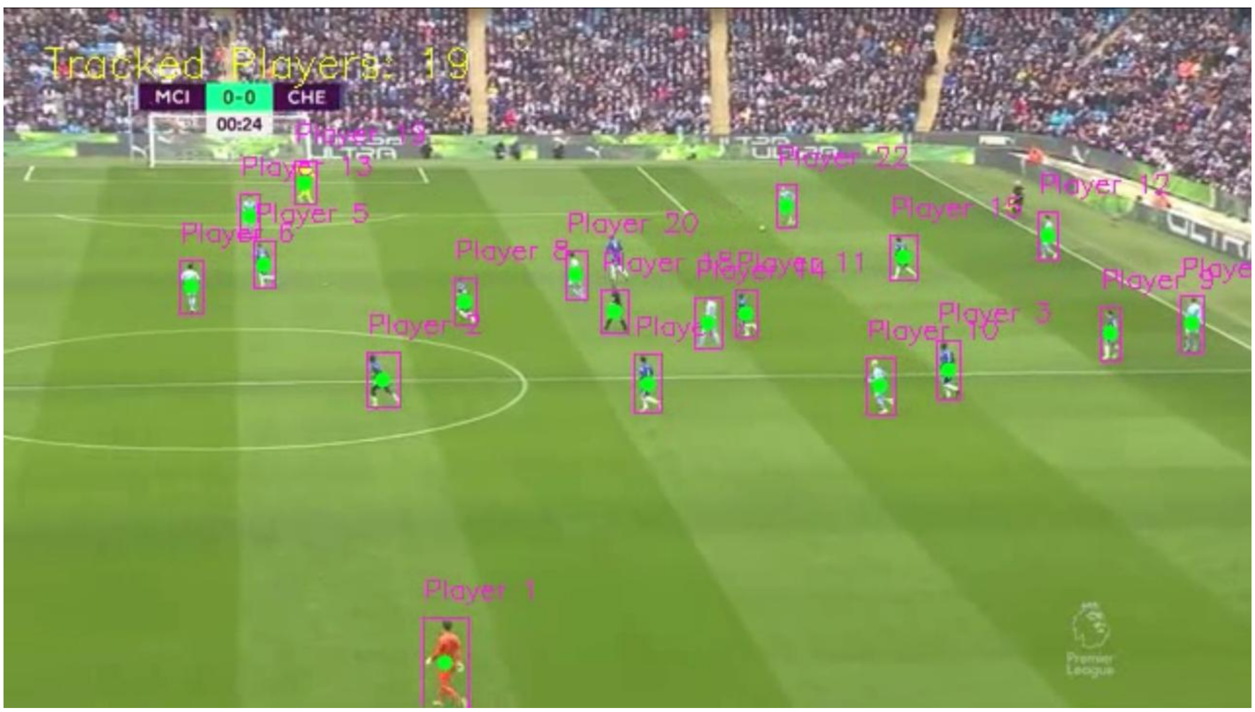

Fig1: - frame1_initial_detection.png

7. Challenges Encountered During Implementation

Several practical challenges were observed while processing real sports footage:

a) Fast Camera Panning:

The tactical football camera continuously moves horizontally to follow gameplay, which causes relative displacement of all tracked players.

b) Dense Player Clustering:

When multiple players move close together, partial overlap between bounding boxes occurs.

c) Similar Team Uniform Appearance:

Players belonging to the same team have similar jersey colors, which occasionally makes identity association more difficult.

d) Partial Occlusion:

Some players are temporarily hidden behind nearby players during running sequences.

These are common real-world challenges in practical multi-object tracking systems.

8. Failure Cases and Observed Limitations

Although the BoT-SORT tracker significantly improves ID persistence, minor ID reassignment was still observed in a few frames where players crossed closely, became very small due to camera zoom, or when the camera panned aggressively.

However, the system still maintains reasonably stable player identities for substantial portions of the processed segment, which satisfies the practical objective of as-stable-as-possible persistent tracking.

Thus, occasional ID switching is treated as a known operational limitation rather than a failure of the complete pipeline.

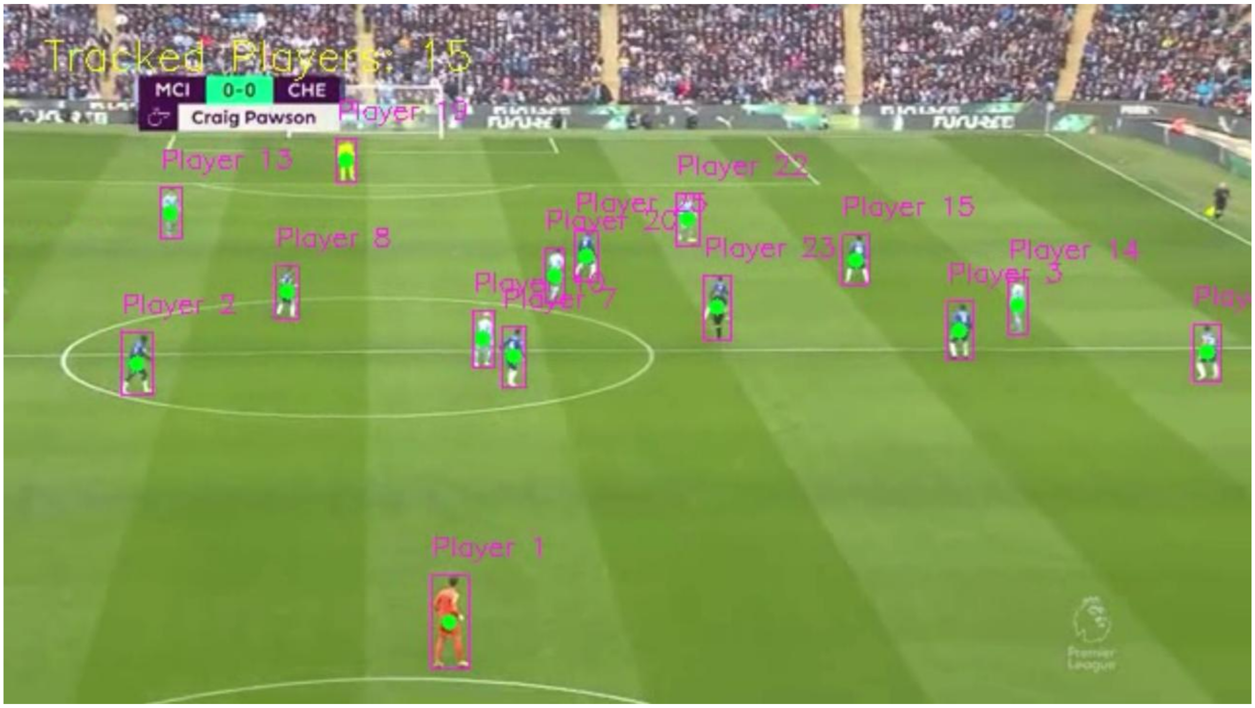

Fig2: - frame2_mid_tracking.png

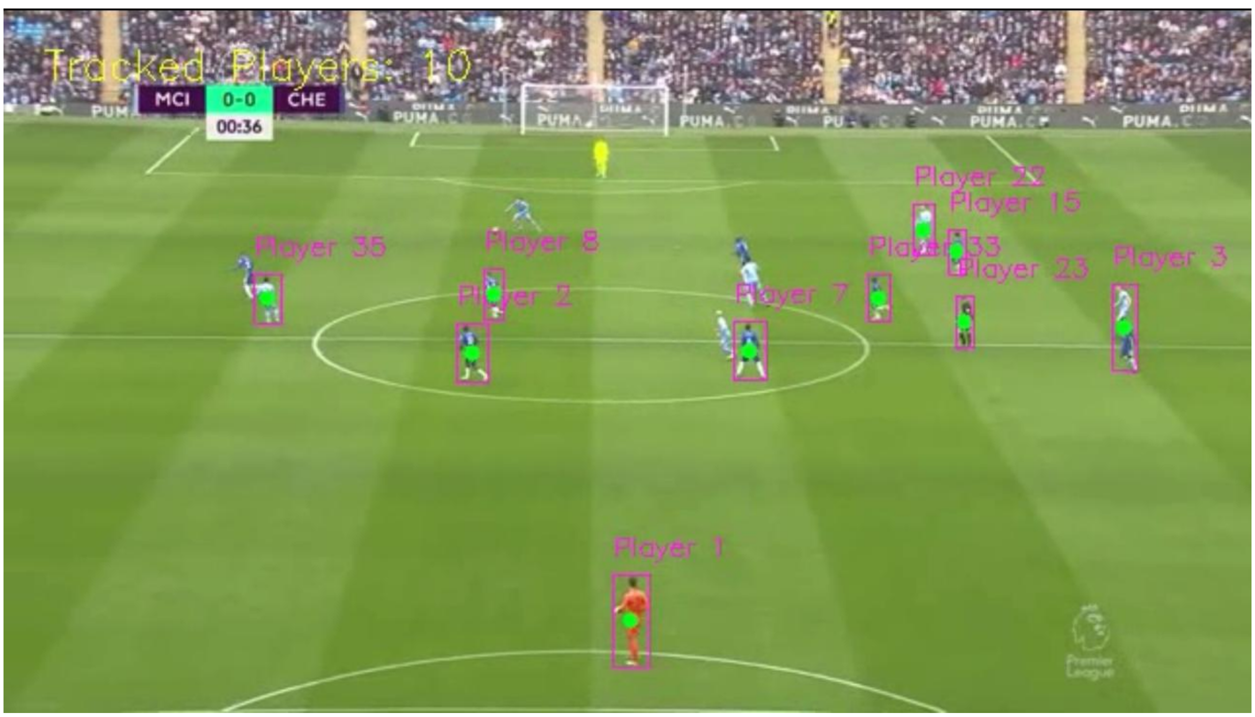

Fig3: - frame3_dense_cluster.png

9. Assumptions Taken

The following assumptions were considered during implementation:

• all visible human subjects in the selected football segment are valid tracking candidates,

• the trimmed 13-second segment is representative enough to demonstrate persistent tracking,

• qualitative visual continuity is sufficient to evaluate tracker performance for this assignment.

10. Future Improvements

The current implementation can be improved further by introducing:

• team jersey color classification,

• player trajectory path visualization,

• stronger person re-identification embeddings,

• speed and motion analytics,

• top-view football field positional mapping.

These additions can make the system more suitable for advanced sports analytics applications.

11. Conclusion

This assignment successfully demonstrates an end-to-end computer vision pipeline for multi-object detection and persistent ID tracking on public football footage using YOLOv8s and BoT-SORT.

The implemented system detects multiple moving players, assigns unique tracking identities, and preserves those identities as consistently as possible while handling practical sports-video challenges such as player overlap, camera motion, and partial occlusion.

Therefore, the required objective of persistent multi-player tracking has been successfully achieved.